

Frank Zhang

Senior Data Engineer

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PROFESSIONAL SUMMARY

Senior Data Engineer with **10 years** of experience designing and scaling enterprise healthcare **data platforms**, specializing in **Azure Databricks Lakehouse** architecture and **Snowflake** enterprise data warehousing. Strong hands-on expertise in building production-grade **ELT** pipelines, dimensional models, and domain-oriented datasets supporting claims analytics, regulatory reporting, and executive **BI**. Proven record of optimizing distributed **Spark** workloads, implementing governed **Snowflake** data marts, and improving pipeline reliability and cost efficiency in **HIPAA**-compliant environments. Experienced mentor contributing to scalable, high-availability data platforms in remote-first teams.

SKILLS

Programming & Query Languages

Python (advanced ETL frameworks, PySpark, data validation), SQL (complex joins, CTEs, window functions, performance tuning), Spark SQL

Big Data & Lakehouse

Azure Databricks, Delta Lake, Medallion Architecture (Bronze/Silver/Gold), Distributed Spark processing, Delta optimization (partitioning, compaction)

Cloud Data Warehousing

Snowflake (RBAC, micro-partition pruning, clustering keys, workload management, secure data sharing), Star Schema, SCD Type 1 & 2, Data Marts

Orchestration & ELT

Apache Airflow (SLA management, retries, backfills, dependency resolution), ELT pipeline design, Incremental processing, CDC-based ingestion

Cloud & Infrastructure

Microsoft Azure (ADLS Gen2, Azure Data Factory, Azure Monitor, Key Vault), Terraform (Infrastructure as Code), CI/CD (Azure DevOps, GitHub Actions)

Streaming (Moderate Exposure)

Apache Kafka (event ingestion), Spark Structured Streaming (incremental updates), near real-time pipeline processing

Data Platform & Governance

Data contracts, metadata standards, environment promotion strategy (Dev/QA/Prod), production support, cost optimization, data observability

Healthcare Domain

HIPAA-compliant architecture, PHI encryption, HL7/FHIR ingestion, Claims 837/835, ICD-10/CPT mapping, audit logging

EXPERIENCE

Elevance Health

Senior Data Engineer

Indianapolis, IN

May 2023— Present

- Led the design and implementation of a scalable Azure Databricks Lakehouse processing over 25TB of claims and clinical data, structuring Bronze, Silver, and Gold layers to improve transformation reliability and reduce analytics turnaround time for actuarial and regulatory reporting teams by 40 percent.
- Engineered distributed PySpark transformation frameworks within Databricks to standardize high-volume 837/835 claims transactions and FHIR-based datasets, improving processing performance and reducing pipeline failures through optimized partitioning and cluster tuning strategies.
- Designed and maintained Snowflake enterprise data marts modeled using Star schema and SCD Type 2 logic to support cost-of-care analytics, provider performance reporting, and CMS regulatory submissions, improving reporting consistency and reducing reconciliation discrepancies.
- Implemented incremental ELT pipelines using Airflow orchestration and CDC-based ingestion patterns to minimize reprocessing overhead and improve data freshness across claims and membership domains.
- Optimized Snowflake warehouse sizing, clustering keys, and micro-partition pruning strategies to reduce compute costs by 30 percent while maintaining high-concurrency performance for BI consumers.
- Built domain-oriented Gold datasets aligned to clinical, financial, and provider analytics teams, improving dataset ownership clarity and reducing cross-team dependency bottlenecks.
- Implemented automated data validation and anomaly detection checks within Databricks workflows to identify claim mismatches, diagnosis inconsistencies, and schema drift, decreasing downstream reporting defects by over 35 percent.
- Developed Terraform-based infrastructure templates to provision Databricks workspaces, storage accounts, and role-based

access policies, improving deployment consistency and reducing environment drift across Dev and Prod.

- Integrated observability metrics within Airflow and Azure Monitor to track pipeline SLA adherence, latency thresholds, and job failure patterns, reducing mean time to resolution for production incidents by 45 percent.
- Supported moderate real-time ingestion requirements by integrating Kafka-based event streams into incremental Spark Structured Streaming workflows for near real-time eligibility updates.
- Collaborated with analytics engineers and BI teams to design Snowflake semantic datasets optimized for Tableau and Power BI dashboards, ensuring consistent KPI definitions across executive reporting.
- Mentored junior engineers on PySpark optimization, Snowflake performance tuning, and production pipeline best practices, improving team velocity and code quality standards.

Cigna

Senior Data Engineer

Bloomfield, CT

May 2017— Apr 2023

- Designed and implemented ELT pipelines ingesting multi-million member claims datasets into ADLS and Databricks for distributed processing, enabling scalable transformation of large healthcare datasets for enterprise analytics initiatives.
- Developed PySpark-based Silver-layer standardization workflows to normalize HL7 and claims data before loading curated Gold datasets into Snowflake for downstream reporting.
- Built Snowflake dimensional models using Star schema and SCD strategies to support provider performance measurement, fraud detection analytics, and financial reporting use cases.
- Orchestrated complex Airflow workflows managing dependency chains, incremental refresh logic, and backfill strategies to ensure reliable execution across Dev and Production environments.
- Transitioned legacy ETL transformations into an ELT pattern by leveraging Snowflake compute for structured SQL transformations, improving maintainability and simplifying lineage tracking.
- Optimized Spark jobs and Snowflake queries by rewriting joins and adjusting partition strategies, significantly improving processing efficiency and reducing reporting latency.
- Implemented HIPAA-compliant encryption, access controls, and audit logging standards across Databricks and Snowflake environments to ensure secure PHI handling and audit readiness.
- Assisted in introducing data contracts and standardized transformation templates to improve dataset consistency across analytics domains.
- Contributed to infrastructure automation efforts using Terraform to provision repeatable Azure storage and analytics environments.
- Supported moderate event-driven ingestion initiatives by integrating Kafka-based transaction feeds into incremental Spark workflows for improved utilization monitoring.
- Built reconciliation checks to validate claims adjudication totals and diagnosis mappings before publishing Gold-layer datasets to Snowflake.
- Collaborated closely with analytics and BI teams to ensure curated Snowflake datasets met semantic and reporting requirements for executive dashboards.

McKinsey

Python Developer

New York, NY

Aug 2015 — Apr 2017

- Developed Python and SQL-based analytics pipelines supporting healthcare payer engagements, transforming operational and claims datasets into structured models for cost optimization initiatives.
- Built dimensional reporting structures to support executive-level utilization and financial forecasting analyses.
- Performed data validation and reconciliation processes to ensure accuracy of client-facing analytical deliverables.
- Integrated disparate claims, membership, and provider datasets into unified reporting models for consistent KPI measurement.
- Optimized SQL transformation logic to improve processing efficiency for large healthcare datasets used in strategy presentations.
- Collaborated with consulting teams to translate business requirements into structured datasets and measurable performance indicators.

EDUCATION

Rice University | Master's Degree in Computer Science | May 2015

Rice University | Bachelor's Degree in Software Engineering | May 2013